Purification of nucleic acids

Risk assessment

– Guanidine salts form TOXIC GASES with bleach or strong acids

▷ Wear gloves, safety glasses, lab coat

□ DO NOT add bleach or acidic solutions directly into sample waste

Reviewed: Jun 10, 2023

This is a bench card. Full protocol available online.

A › *Purification by alcohol precipitation*

(1.) Add a precipitating salt to the sample at the final concentration listed below; mix well. Compute the spike volume from the available stock.

Sample contains	Salt	Final	Caveat
DNA or RNA (default)	NaOAc	0.3 M	
Long RNA (selective)	LiCl	0.8 M	not compatible with in vitro translation or reverse transcription
dNTPs or oligosaccharides	NH_4OAc	2.0–2.5 M	not compatible with T4 PNK
SDS	NaCl	0.2 M	

(2.) *Optional:* Add 10–20 µg/mL linear polyacrylamide to precipitate picogram quantities of nucleic acids.

(3.) Add 2.0 vol ethanol to precipitate DNA, or 2.5–3.0 vol ethanol to precipitate RNA. Mix by vortexing.

(4.) Precipitate for 15–30 min on ice.

(5.) Centrifuge at 12 000 × *g* for 15 min at 4 °C. Discard supernatant and spin briefly to remove residual solvent. The pellet will appear nearly transparent.

(6.) Wash pellet with 650 µL Buffer PE or freshly prepared 70% ethanol. Discard supernatant carefully.

(7.) Dry pellet for 10–15 min at room temperature, or for 1 min in a vacuum centrifuge.

Critical: Avoid overdrying, which impairs dissolution and may denature small dsDNA.

(8.) Resuspend in 15–30 µL 5 mM Tris pH 8.0 or buffer of choice.

B › *Purification over a silica matrix*

(1.) Mix 1.0 vol sample with 2.0 vol Buffer PB.

(2.) Apply up to 700 µL to a silica spin column. Centrifuge at 11 000 × *g* for 30 s or use vacuum manifold.

(3.) Wash with 650 µL Buffer PE.

Critical: Let stand for 2–5 min before spinning if DNA will be used in salt-sensitive applications such as blunt-end ligation.

(4.) Repeat wash to minimize salt carryover.

(5.) Dry membrane by centrifugation at 11 000 × *g* for 1 min.

Critical: Residual ethanol may persist if flow-through is not discarded prior to drying.

(6.) Place spin column in a clean tube. Apply 15–30 µL 5 mM Tris pH 8.0 onto the silica membrane. Incubate 1 min. Elute at 11 000 × *g* for 1 min.

(7.) *Optional:* Repeat elution using the same or half the volume to improve yield. Pool eluates if desired.

Recipe (available online) Troubleshooting (available online) Notes (available online)

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